

Look Before You Act: Enhancing Vision Foundation Representations for Vision-Language-Action

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Project page: <https://deepvision-vla.github.io/>

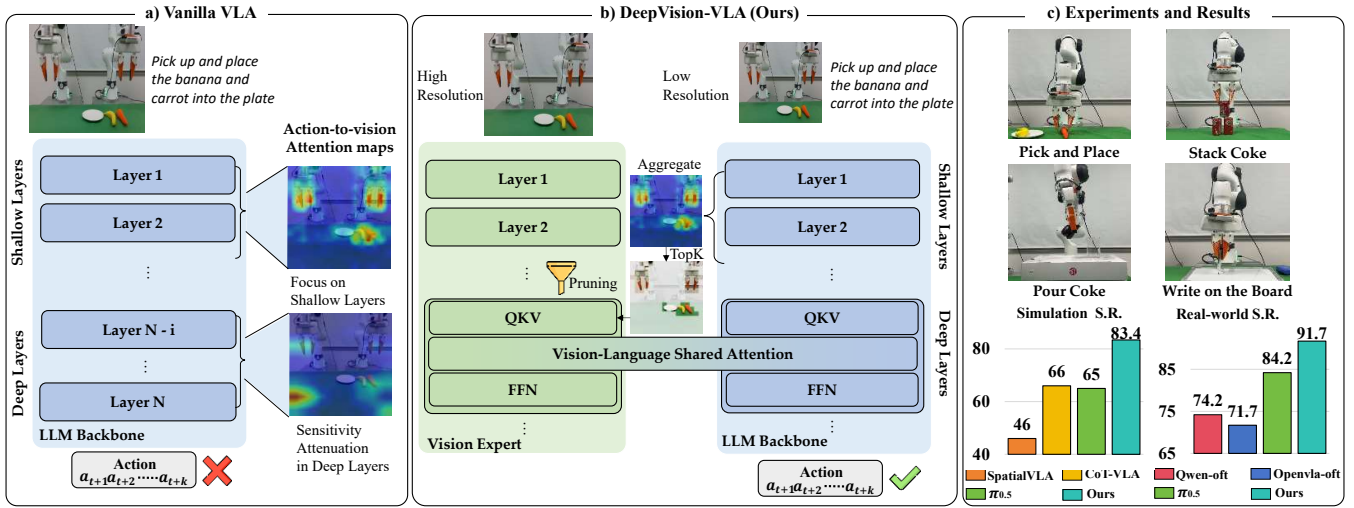


Figure 1: Overview. (a) In vanilla VLA models, action-to-vision attention progressively weakens in deeper layers, leading to visual drift and reduced sensitivity to task-relevant objects. (b) DeepVision-VLA employs a VL-MoT framework that uses action-guided saliency from shallow layers to select high-resolution visual features from a Vision Expert and fuse them into deeper LLM layers, reinforcing visual grounding. (c) Under this paradigm, DeepVision-VLA achieves superior performance in both simulated and real-world tasks.

Abstract

Vision-Language-Action (VLA) models have recently emerged as a promising paradigm for robotic manipulation, in which reliable action prediction critically depends on accurately interpreting and integrating visual observations conditioned on language instructions. Although recent works have sought to enhance the visual capabilities of VLA models, most approaches treat the LLM backbone as a black box, providing limited insight into how visual information is grounded into action generation. Therefore, we perform a systematic analysis of multiple VLA models across different action-generation paradigms and observe that sensitivity to visual tokens progressively decreases in deeper layers during action generation. Motivated by this observation, we propose **DeepVision-VLA**, built on a **Vision-Language Mixture-of-Transformers (VL-MoT)** framework. This framework enables shared attention between the vision foundation model and the VLA backbone, injecting multi-scale visual features from the vision expert into deeper layers of the

VLA backbone to enhance visual representations for precise and complex manipulation. In addition, we introduce **Action-Guided Visual Pruning (AGVP)**, which leverages shallow-layer attention to prune irrelevant visual tokens while preserving task-relevant ones, reinforcing critical visual cues for manipulation with minimal computational overhead. DeepVision-VLA outperforms prior state-of-the-art methods by xx% and xx% on simulated and real-world tasks, respectively, providing new insights for the design of visually enhanced VLA models.

1 Introduction

Driven by training on massive, internet-scale multimodal corpora [1, 26, 28, 29, 42], recent advances in Vision-Language Models (VLMs) have demonstrated exceptional proficiency in perception, reasoning, and instruction following [2, 4, 12, 22, 39]. Capitalizing on these strengths, Vision-Language-Action (VLA) models extend

these capabilities to robotics by directly mapping multimodal observations and natural language instructions to robot actions. Powered by billions of parameters and large-scale robot pre-training datasets [23, 40, 55], VLAs have exhibited remarkable potential for learning generalizable manipulation skills across diverse tasks and environments.

The robust control and generalization of VLA models are fundamentally contingent upon the precise interpretation and integration of visual observations [59]. Consequently, contemporary research has increasingly focused on fortifying the visual understanding and reasoning of VLAs. Existing approaches enhance the visual capabilities of VLA models from four main perspectives: (1) introducing visual prompts to improve the model’s understanding of scenes and manipulated objects [18, 31, 49]; (2) designing auxiliary visual objectives to encourage the model to focus on task-critical entities [21, 46]; (3) incorporating additional visual modalities to provide complementary information [27, 49, 57, 61]; and (4) predicting future states to strengthen the model’s ability to model the physical world [36, 37, 60]. Despite these advancements, existing paradigms typically treat the underlying Large Language Model (LLM) backbone in VLA models as a monolithic “black box”, offering little insight into how visual information is propagated and utilized.

In this work, we move beyond treating the LLM backbone in VLA models as an opaque module and instead investigate how visual information is processed across its internal layers. Since the LLM is composed of stacked Transformer layers, a layer-wise analysis offers a natural and granular lens for understanding multimodal integration. We conduct a systematic investigation of several representative VLA architectures and action generation paradigms in two complementary stages. First, we qualitatively analyze action-to-visual attention maps and observe that while early layers maintain grounded attention on task-relevant objects, deeper layers often fail to focus effectively on these regions. To quantify the impact of this phenomenon on action generation, we introduce a layer-wise visual token dropout strategy. The results align with our qualitative findings: in deeper layers, action accuracy becomes increasingly insensitive to the masking of task-relevant visual regions, whereas shallow layers exhibit the opposite trend. We attribute this pattern to the prevalent serial architecture of current VLA models, where visual information is injected only at the first LLM layer and gradually attenuates as it propagates through deep Transformer layers.

Motivated by these findings, we introduce **Vision-Language Mixture-of-Transformers (VL-MoT)**—a novel VLA framework designed to enhance the sensitivity of deeper layers to task-relevant visual regions for improved action prediction. In addition to the original visual encoder, our framework incorporates a dedicated **Vision Expert** (DINOv3, 0.8B) whose representations are fused with the VLA backbone via a MoT mechanism. Building on our observation that shallow layers naturally maintain effective visual grounding, we adopt an integration scheme that selectively couples the Vision Expert only with deeper VLA layers. Specifically, we extract multi-scale visual features by sampling from the last few Transformer layers of the Vision Expert. This strategy empirically outperforms alternative integration approaches, such as sampling

from the early layers or uniformly across all layers. A plausible explanation is that the later layers of the Vision Expert capture higher-level, semantically rich representations that are more invariant and object-centric, making them more compatible with task-relevant, action-conditioned features in the VLA model. Through this targeted integration, the Vision Expert collaborates with deeper VLA layers to reinforce action generation precisely where the model is most susceptible to visual drift, ultimately enabling more reliable and visually-anchored robotic control.

However, naively integrating the full feature maps from the Vision Expert into the VLA backbone may introduce significant redundancy and irrelevant background information, potentially diluting task-critical signals. To address this, we propose an **Action-Guided Visual Pruning (AGVP)** strategy that refines information flow within our framework. Specifically, we leverage the robust grounding capabilities of shallow VLA layers to compute a saliency map by averaging attention weights from action tokens to visual tokens. This map identifies the most task-relevant regions, which we then use to prune the Query, Key, and Value across the Vision Expert’s Transformer layers before integrating them into the deep VLA layers. This targeted pruning not only mitigates visual redundancy but also enables the Vision Expert to process higher-resolution inputs with minimal computational overhead. Our empirical results demonstrate that this increased visual granularity, which focuses precisely on action-critical entities, leads to a significant boost in manipulation performance.

We instantiate our framework as **DeepVision-VLA**, built upon a custom baseline, QwenVLA-OFT. This baseline leverages a Qwen3-VL backbone (4B) and adopts parallel action decoding with L1 regression output [24]. We systematically evaluate DeepVision-VLA across ten simulated tasks in RL-Bench [20] and four complex dual-arm real-world manipulation tasks. DeepVision-VLA achieves state-of-the-art (SOTA) performance, outperforming prior VLA methods by 8% in simulated settings and 13% in real-world settings. Our contributions are summarized as follows:

- We systematically analyze visual information utilized in current VLA models and identify a phenomenon where deeper LLM backbone become insensitive to task-relevant visual regions.
- We propose Vision-Language Mixture-of-Transformers (VL-MoT), a novel framework that improves action prediction by injecting multi-scale visual features from a Vision Expert into deep VLA layers.
- We introduce Action-Guided Visual Pruning, a strategy that filters redundant visual information from the Vision Expert, providing deeper VLA with task-relevant visual signals.
- DeepVision-VLA establishes SOTA results in both simulation and real-world settings, demonstrating the effectiveness of our framework and providing more insights for the design of visually enhanced VLA models.

2 Related Work

Vision-Language-Action (VLA) Models. VLA models [3, 5, 6, 11, 19, 25, 33, 34, 53, 54] are primarily driven by scaling robot demonstration data [8, 23, 40, 55] and adapting pretrained vision-language models (VLMs) [2, 12, 22, 39] for robotic control. These approaches

directly model action sequences from visual observations and language instructions, demonstrating strong scalability and significant potential for generalization. Early work attempted to leverage the autoregressive capabilities of pretrained VLMs to generate robot actions token by token [3, 7, 25, 62]. However, such formulations often suffer from action discontinuities and low execution frequency. Inspired by the success of diffusion policies [13, 58], recent efforts have explored diffusion-based [30, 33, 54] and flow-based VLA [6, 19] frameworks, which leverage the strong representation power of VLMs while introducing a dedicated action head to learn smooth and stable continuous action outputs. To further improve execution efficiency, several works [9, 11, 14] adopt a dual-system design, where a reasoning module is responsible for task planning, while a control module focuses on action generation. In addition, hierarchical architectures [43, 43] have been proposed to better handle high-level and abstract human instructions, typically leveraging an auxiliary VLM to decompose complex instructions into subgoals that guide the VLA for downstream instruction-following action generation. However, recent studies [21, 46, 59] suggest that when VLA models do not sufficiently develop their visual understanding during training, their action modeling performance can degrade noticeably, which may hinder precise manipulation in dynamic or cluttered environments. These findings suggest that preserving reliable, task-aware visual representations can be crucial for effective action generation [15, 50, 56].

Vision Improvement for VLA. As precise action generation critically depends on robust visual understanding and grounding [46, 59], a growing body of work has explored strategies to enhance VLA models from a visual perspective. One line of research augments the VLA input with additional visual cues to facilitate task comprehension, such as overlaying execution trajectories [18, 31] or highlighting target objects [17], demonstrating that simple prompt engineering can be surprisingly effective. Another approach introduces auxiliary visual supervision to encourage the model to attend to important image regions, for example by reconstructing key objects in image [46] or anchoring the VLA’s visual representations to strong teacher features [21], thereby improving the reliability of action generation. Beyond 2D inputs, incorporating richer visual modalities such as depth maps [32, 51, 57], 3D point clouds [27, 41, 61] or hand-drawn sketch [49] provides complementary spatial and geometric information, enabling the model to better reason about object shapes, distances, and occlusions. Finally, several methods adopt a reasoning-before-action paradigm [10, 36, 60], predicting future states or images to strengthen the model’s understanding of physical dynamics, thereby enhancing the accuracy of manipulation. Despite these advancements, existing paradigms often treat the visual processing within VLA models as a black box, focusing primarily on the input state and the resulting actions while largely overlooking how visual information is internally utilized. In this work, we provide insights into this process and propose DeepVision-VLA, which integrates multi-scale features from a visual foundation model into the deeper layers of the VLA via an Vision-Language Mixture-of-Transformers architecture, enhancing attention to task-relevant objects and improving action generation accuracy.

3 Methods

In this section, we first introduce the problem formulation and the general architecture of Vision-Language-Action (VLA) models in Sec. 3.1. Next, Sec. 3.2 analyzes how representative VLA architectures process and utilize visual information, providing key insights into their limitations. Building on this analysis, we propose the **Vision-Language Mixture-of-Transformers (VL-MoT)** framework, which improves action prediction performance by injecting multi-scale Vision Expert knowledge into the deep layers of VLA models. Sec. 3.3 presents **DeepVision-VLA**, a concrete instantiation of this framework, and introduces the **Action-Guided Vision Pruning (AGVP)** strategy to further enhance the model’s focus on task-relevant visual regions. The overall framework is shown in Figure 3. Finally, Sec. 3.4 describes the training and inference procedures of DeepVision-VLA.

3.1 Preliminaries

Problem Formulation. We consider a standard imitation learning setting for vision-language robotic manipulation. Given a dataset of expert demonstrations $\mathcal{D} = \{(\tau_i, l_i)\}_{i=1}^N$, where l_i denotes the language instruction and $\tau_i = \{(o_t, a_t)\}_{t=1}^{T_i}$ represents the corresponding trajectory consisting of visual observations and actions, the goal is to learn a policy π_θ that maps visual observations and task instructions to robot actions. At each time step t , the policy takes the current observation o_t together with the instruction l as input and predicts robot actions. Depending on the action representation, the policy may either predict the current action a_t or a short horizon of future actions $a_{t:t+H-1}$ under an action chunking formulation. Formally, this can be written as $\pi_\theta(a_t \mid o_{\leq t}, l)$ or $\pi_\theta(a_{t:t+H-1} \mid o_{\leq t}, l)$. The policy parameters are learned from demonstrations by solving

$$\theta^* = \arg \min_{\theta} \mathbb{E}_{(\tau, l) \sim \mathcal{D}} \left[\sum_{t=1}^T \ell(\pi_\theta(o_{\leq t}, l), a_t) \right], \quad (1)$$

where $\ell(\cdot, \cdot)$ denotes the task-specific action supervision objective.

Vision-Language-Action Models. A typical VLA model consists of three primary components: a visual encoder, an LLM backbone, and an action decoder. The visual encoder extracts visual features from input observations, while the LLM backbone integrates these visual representations with the language instruction to perform multimodal reasoning and action conditioning. Finally, the action decoder maps the resulting representations to the robot action space for execution. Existing VLA models mainly differ in how actions are represented and learned. For instance, OpenVLA formulates action generation as *next-token prediction*, where discretized actions are generated autoregressively. In contrast, π_0 adopts *flow matching* to model continuous actions, while OpenVLA-OFT employs *parallel decoding*, predicting multiple future actions simultaneously using bidirectional attention for efficient chunk-level prediction. Despite these differences, all approaches rely on the LLM’s ability to effectively interpret both visual observations and language instructions, which is critical for accurate action prediction. In this work, we focus on investigating how visual information influences the action prediction performance of VLA models.

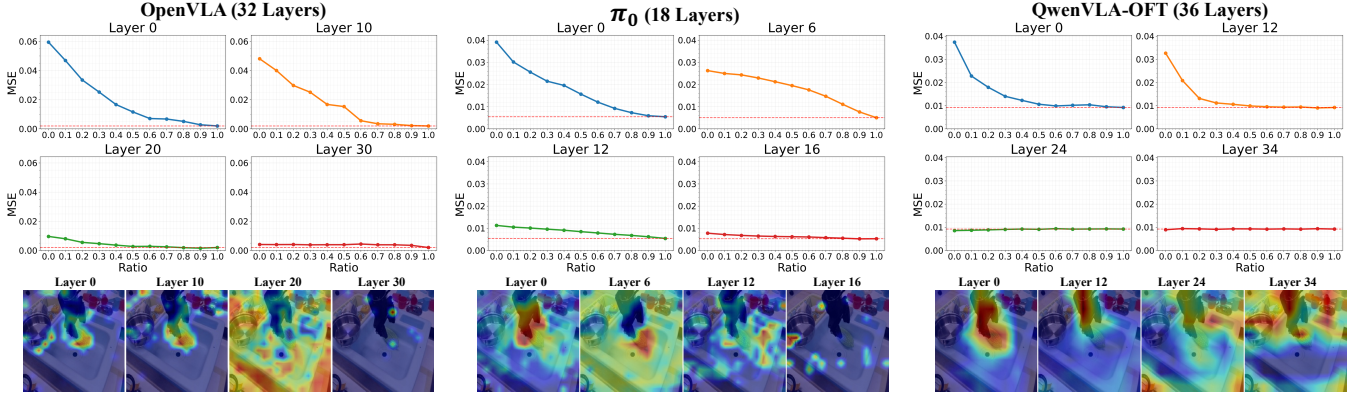


Figure 2: Layer-wise analysis of visual grounding in VLA models. Top: Effect of masking ROI visual tokens on action prediction error (MSE) at different layers. Masking in shallow layers significantly degrades performance, while the impact diminishes in deeper layers. Bottom: Action-to-vision attention maps across layers. Attention is concentrated on task-relevant regions in shallow layers but becomes increasingly diffuse in deeper layers, indicating weakened visual grounding.

3.2 Probing the Role of Vision in VLA Models

Given that the LLM backbone in VLAs is composed of stacked Transformer layers, we conduct a layer-wise investigation to understand how visual information is processed throughout action prediction. Specifically, we first analyze action-to-vision attention maps to determine whether the model effectively attends to task-relevant objects in the scene, as such grounding provides a measure of reliable action learning. Next, to quantitatively evaluate the contribution of visual tokens to action performance, we perform a controlled, layer-wise ablation by masking critical visual tokens and observing the corresponding changes in action prediction accuracy. In the following, we describe our experimental setup and present the observed results along with their analysis.

Experimental Setup. We evaluate three representative VLA models that differ in their LLM backbones, model depths, and action generation paradigms: OpenVLA [25], π_0 [6], and a custom baseline QwenVLA-OFT, which adopts Qwen3-VL [2] as the backbone and performs parallel action prediction with an ℓ_1 regression objective. Our analysis is conducted on 1,500 randomly sampled trajectories from the BridgeV2 [52] dataset, which offers high-quality manipulation demonstrations with clear object layouts and consistent visual observations, making it particularly suitable for studying object-level visual sensitivity. For each image, we employ Grounding-DINO [35] to localize the regions of interest (ROIs), including the robot arm, the manipulated object, and their interaction area.

Action-to-Vision Attention. To better understand how VLA models ground action predictions in visual information, we visualize the attention from each action token to all visual tokens across LLM layers. At each layer, we average the attention weights from all action tokens to the visual tokens, producing a layer-wise action-to-vision attention map that captures how strongly the model relies on visual information at that depth. As shown in Figure 2 (bottom), all three VLA paradigms exhibit a consistent pattern: in shallow layers, action tokens attend predominantly to the visual ROIs localized for the task, including the manipulated object and the robot arm. In contrast, in deeper layers, attention becomes increasingly diffuse and extends to less relevant regions, indicating that the grounding

of action tokens in visual information progressively diminishes along the LLM backbone.

Action Prediction Sensitivity to ROI Visual Tokens. While attention visualization provides qualitative insights into how models attend to visual regions, it does not directly quantify how much action prediction actually depends on those regions. To more rigorously measure the contribution of task-relevant visual information, we perform a layer-wise masking study on visual tokens corresponding to the ROIs. Specifically, for a selected layer in LLM, we identify the visual tokens associated with the ROIs and zero out a fraction r of them, effectively removing their information from the model, while keeping all non-ROI tokens unchanged. The resulting hidden states are then propagated through the remaining layers to produce the final action prediction. We measure the effect of this intervention using the mean squared error (MSE) between the predicted actions and the ground-truth actions. Since the masking is restricted to ROI tokens at a specific layer, the resulting change in performance directly reflects the extent to which grounded visual information contributes to action prediction at that depth. As shown in Figure 2 (top), all three models exhibit a consistent layer-wise pattern. Masking ROI tokens in early layers leads to a substantial increase in action MSE, indicating that these layers rely heavily on task-relevant visual information. In contrast, the effect of masking progressively decreases in deeper layers, and even complete removal of ROI tokens in the deepest layers causes only minor changes in action prediction. These results indicate that task-relevant visual cues become increasingly underutilized in deep layers, which may partially undermine the reliability of the learned actions.

3.3 DeepVision-VLA

Based on the above analysis and the insight that increasing the sensitivity of deep VLA layers to task-relevant visual ROIs may improve action prediction accuracy, we propose the Vision-Language Mixture-of-Transformers (VL-MoT) framework. To facilitate understanding, we illustrate how the framework is applied to the QwenVLA-OFT baseline to construct DeepVision-VLA, as shown in Figure 3 (a). In this section, we begin by introducing the baseline

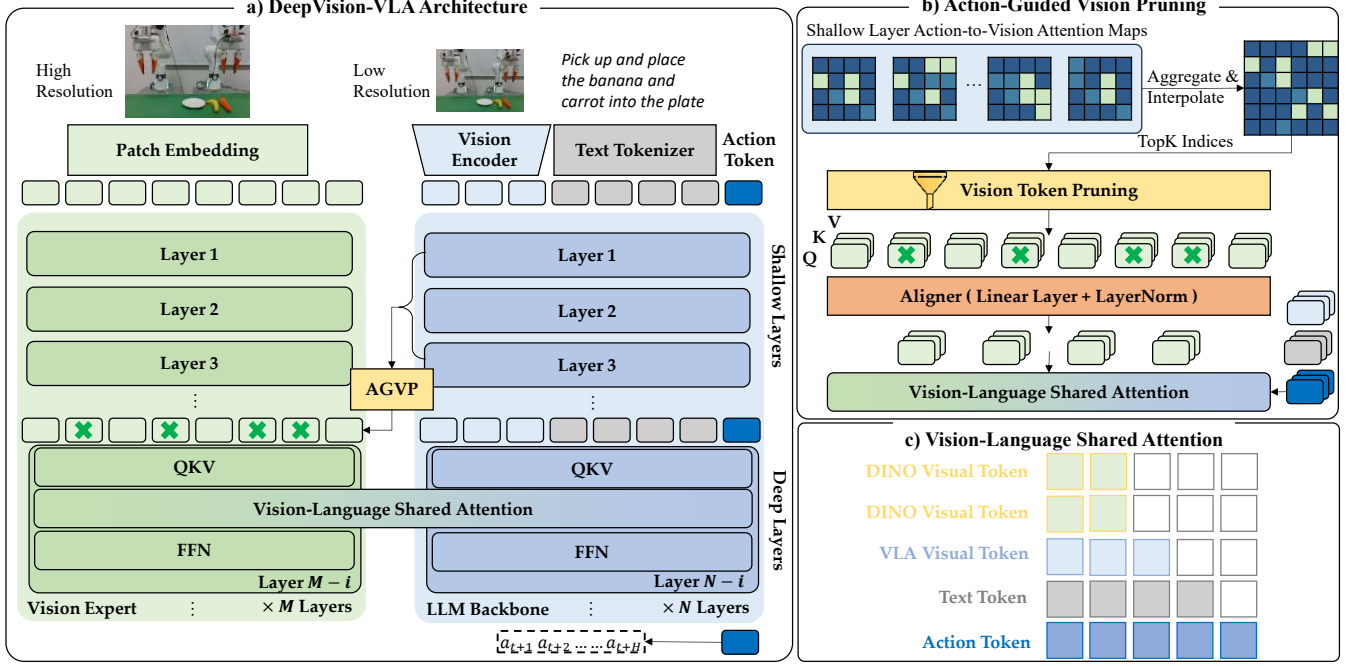


Figure 3: Framework. (a) A high-resolution Vision Expert is coupled with the LLM backbone through the proposed Vision–Language Mixture-of-Transformers (VL-MoT) framework, where deep LLM layers share attention with the Vision Expert to enhance visual grounding for action prediction. (b) Action-to-vision attention from shallow LLM layers is aggregated to identify task-relevant regions, which are used to prune Vision Expert tokens before fusion. (c) Vision Expert tokens use bidirectional attention to preserve its pretrained knowledge. VLA tokens apply causal attention to prompts and bidirectional attention to action tokens for parallel prediction.

model, followed by the architectural design of AVL-MoT, and finally present the Action-Guided Vision Pruning (AGVP) strategy.

3.3.1 QwenVLA-OFT Baseline. The baseline model is built upon Qwen3-VL (4B) [2] and adopts the parallel action prediction paradigm with an ℓ_1 regression objective as proposed in OpenVLA-OFT [24]. It consists of three main components: a visual encoder, an LLM backbone, and an action decoder.

Visual Encoder. QwenVLA-OFT utilizes SigLIP2-Large (0.3B) as its visual encoder to extract expressive image representations. The encoder employs 2D-RoPE with interpolated absolute positional embeddings to provide spatial information for visual tokens. Given an input image $I \in \mathbb{R}^{H \times W \times 3}$, it is partitioned into patches to produce dense visual features. To reduce token count and improve computational efficiency, neighboring 2×2 features are merged into a single token and projected to the LLM hidden dimension d via a two-layer MLP, resulting in $\mathbf{V} = E_{\text{vis}}(I) \in \mathbb{R}^{N_v \times d}$, where N_v is the number of visual tokens.

LLM Backbone. The visual features $\mathbf{V}$ are concatenated with tokenized language instructions and action tokens initialized as zero vectors to form the multimodal input sequence to the LLM. At layer ℓ , the hidden states are decomposed as $\mathbf{Z}^\ell = [\mathbf{Z}_V^\ell; \mathbf{Z}_L^\ell; \mathbf{Z}_A^\ell]$, where $\mathbf{Z}_V^\ell$, $\mathbf{Z}_L^\ell$, and $\mathbf{Z}_A^\ell$ denote the visual, language, and action embeddings, respectively. The LLM updates these hidden states layer by layer according to $\mathbf{Z}^\ell = \mathcal{F}^\ell(\mathbf{Z}^{\ell-1})$ for $\ell = 1, \dots, L$, and the final action embedding $\mathbf{Z}_A^L$ is then fed to the action decoder.

Action Decoder. The action decoder maps the final action embeddings $\mathbf{Z}_A^L$ to the robot’s action space using a two-layer MLP. Specifically, the predicted action vector $\mathbf{a} \in \mathbb{R}^n$, where n is the number of degrees of freedom (DoF) of the embodiment.

3.3.2 VL-MoT Framework. To enhance visual grounding in the deeper layers of the VLA, we introduce a **Vision Expert**, DINOv3. Its multi-scale transformer features are injected into deep VLA layers via a shared-attention mechanism. This design allows the VLA to incorporate multi-scale visual information into layers that would otherwise be insensitive to task-relevant visual ROIs, thereby improving the precision of action generation.

Vision Expert. We adopt the visual foundation model DINOv3 [45] because it provides spatially detailed representations [24, 25, 30] that are more fine-grained than the high-level semantic features typically extracted by existing VLA visual encoders, making it particularly suitable for precise manipulation tasks. Our approach, however, is not limited to this choice, and exploring alternative vision experts is left for future work.

VL-MoT Design. To integrate the multi-scale knowledge of the Vision Expert into the deep VLA layers, we do not simply concatenate intermediate features and process them jointly through the LLM. Instead, we introduce a novel MoT architecture that directly exposes the intermediate Query, Key, and Value (QKV) representations from the Vision Expert and integrates them with the corresponding QKV of the deep VLA layers via a shared-attention mechanism, as shown in Figure 3 (a). This enables cross-branch

information exchange while preserving separate processing pathways, effectively reducing feature interference and stabilizing fusion training.

To align with the layers where visual grounding becomes less sensitive, the Vision Expert is connected only to the deepest n layers of the VLA, where n is determined based on the analysis in Sec. 3.2. For the Vision Expert, we select the last n transformer layers and use their QKV representations for feature injection. To formally describe this interaction, we consider a coupled layer between the Vision Expert and LLM backbone. Here, $\mathbf{E}^k \in \mathbb{R}^{M \times d_e}$ represents the features from the k -th transformer layer of the Vision Expert, where M is the number of expert tokens and d_e is the expert hidden dimension. Similarly, $\mathbf{Z}^\ell \in \mathbb{R}^{N \times d}$ denotes the token representations of the LLM at layer ℓ , where N is the number of LLM tokens and d is the LLM hidden dimension. We then compute the QKV projections for the Vision Expert and the LLM tokens separately:

$$\begin{aligned} Q_E &= \mathbf{E}^k W_Q^E, & K_E &= \mathbf{E}^k W_K^E, & V_E &= \mathbf{E}^k W_V^E, \\ Q_Z &= \mathbf{Z}^\ell W_Q^Z, & K_Z &= \mathbf{Z}^\ell W_K^Z, & V_Z &= \mathbf{Z}^\ell W_V^Z. \end{aligned} \quad (2)$$

Since the hidden dimensions of the two branches may differ, the QKV representations of the Vision Expert tokens are each projected through a learnable linear layer to align with the LLM hidden dimension. The shared attention is then computed by concatenating the QKV representations from both branches:

$$Q = [Q_E; Q_Z], \quad K = [K_E; K_Z], \quad V = [V_E; V_Z]. \quad (3)$$

$$A = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right), \quad H = AV. \quad (4)$$

Finally, we split the output $H = [H_E; H_Z]$ back into the two branches and apply the remaining standard Transformer operations. Notably, we maintain the original bidirectional attention mechanism for visual expert tokens, as shown in Figure 3 (c), which better preserves pre-trained knowledge and provides more stable visual features for the VLA model.

Empirically, this VL-MoT configuration outperforms alternative feature selection strategies, such as using QKV features exclusively from the first n layers or uniformly sampling n layers across the entire Vision Expert. We attribute this design choice to the fact that the deeper layers of the Vision Expert encode high-level, semantically rich, and object-centric representations, which align naturally with the task-relevant, action-conditioned features in the VLA model and are particularly effective for fine-grained and complex manipulation.

3.3.3 Action-Guided Vision Pruning. To enable the VLA model better focus on task-relevant objects, we propose Action-Guided Visual Pruning, as shown in Figure 3 (b), which filters redundant background features before integrating multi-scale information from the Vision Expert into deeper VLA layers.

Based on our observation in Section 3.2 that shallow VLA layers provide reliable action-conditioned visual grounding, we use the attention from action tokens to visual tokens in these layers to identify ROI visual tokens. Let $\mathbf{A}^\ell \in \mathbb{R}^{N_a \times N_v}$ denote the action-to-vision attention map at layer ℓ , where N_a and N_v are the numbers of action and visual tokens, respectively. Since multiple action tokens jointly contribute to robot control, we first average their attention as

first average over the action tokens as $\mathbf{m}^\ell = \frac{1}{N_a} \sum_{i=1}^{N_a} \mathbf{A}_{i,:}^\ell$, to obtain a per-layer attention map, and then aggregate over a set of shallow layers $\mathcal{L}_s$ to obtain the final attention map $\mathbf{m} = \frac{1}{|\mathcal{L}_s|} \sum_{\ell \in \mathcal{L}_s} \mathbf{m}^\ell$. This multi-layer averaging effectively filters out noise and produces stable task-relevant ROI visual tokens.

Compared with the input image used in the VLA, we leverage the strong feature extraction capability of the Vision Expert by feeding a higher-resolution image, allowing the model to capture more fine-grained object details. Consequently, we interpolate the obtained attention map to match the resolution of the Vision Expert. Denoting the interpolation operator by $\mathcal{I}(\cdot)$, the transferred attention map is $\tilde{\mathbf{m}} = \mathcal{I}(\mathbf{m}) \in \mathbb{R}^{N_d}$, where N_d is the number of Vision Expert image tokens. Since this attention map effectively indicates the importance of each region, we retain the top- K tokens with the highest importance, whose indices are given by $\mathcal{S}_K = \text{TopK}(\tilde{\mathbf{m}}, K)$, and the corresponding tokens for sharing attention with deep VLA layers are $\tilde{\mathbf{E}}^k = \mathbf{E}^k[\mathcal{S}_K]$. This approach not only effectively injects critical visual region information into deep VLA layers but also controls the computational budget while exploiting the high-resolution input of the Vision Expert.

3.4 Training and Inference

Training Recipe. Prior to pretraining DeepVision-VLA, we initialize the model with pretrained weights from Qwen3-VL [2] and DINOv3 [45], and then train the entire architecture in an end-to-end manner. Following prior work [11, 17, 33, 36], we curate a specialized pretraining dataset by carefully processing and filtering several large-scale cross-embodiment datasets, including Open X-Embodiment [40], DROID [23], and RoboMIND [55]. As detailed in the Appendix, the resulting dataset contains over 400K trajectories, and DeepVision-VLA is trained on this dataset for one epoch. The fine-tuning settings for downstream tasks are described in the experimental section, as they differ between simulation and real-world environments.

Inference. At inference time, DeepVision-VLA first takes the current image observation and language instruction as input. These inputs are propagated through the shallow layers of the VLA, where action-to-vision attention maps are computed to identify task-relevant visual regions. Based on these cues, we prune the multi-scale features from the Vision Expert and integrate them into the deeper VLA layers through the VL-MoT mechanism. Finally, after the deep layers complete the forward pass, the hidden states of the action tokens are fed into the action decoder to generate the final actions. Notably, this procedure for identifying key visual regions and enhancing VLA perception requires no additional annotations or external supervision, and the entire pipeline remains end-to-end executable after training.

4 Experiments

In Section 4.1, we systematically evaluate the manipulation capability of our proposed DeepVision-VLA in simulated environments. Subsequently, Section 4.2 presents an ablation study to validate the essential roles of visual foundation experts and AGVP strategy. Finally, in Section 4.3, we further demonstrate the model’s effectiveness through single-arm real-world manipulation tasks.

4.1 Simulation Experiment

Data Collection. We evaluate DeepVision-VLA across 10 manipulation tasks from the RLBench [20] benchmark based on the CoppeliaSim simulation environment, including 1) *Close box*, 2) *Close Laptop*, 3) *Toilet seat down*, 4) *Sweep to dustpan*, 5) *Close fridge*, 6) *Phone on base*, 7) *Take umbrella out*, 8) *Frame off hanger*, 9) *Wine at rack*, and 10) *Water plants*. All tasks are performed on a Franka Panda robot with front-view RGB observations. Following pre-defined waypoints and utilising the Open Motion Planning Library [48], we construct a training dataset in which each task comprises 100 trajectories, with the frame-sampling technique used in previous studies [44].

Training and evaluation protocol. We compare DeepVision-VLA with 6 representative Vision-Language-Action (VLA) baselines including OpenVLA [25], SpatialVLA [41], CogACT [30], CoT-VLA [60], $\pi_{0.5}$ [19] and HybridVLA [33]. Each baseline is initialised with its officially released weights and then fine-tuned with all parameters, strictly adhering to the configurations and hyperparameters recommended in its original publication. To ensure a rigorous and fair comparison, we modify OpenVLA-OFT by integrating DINOv3 visual features into its deep vision-degraded layers, consistent with the architectural enhancement strategy employed in our proposed framework. Regarding input modalities, DeepVision-VLA processes multi-modal observations comprising 384×384 resolution RGB images, language instructions provided by the simulation environment, and 7-DoF robotic proprioceptive data. We train DeepVision-VLA through two distinct stages with 1 epoch for pretraining and 300 epochs for downstream tasks, respectively, using the AdamW [38] optimizer on 8 NVIDIA H20 GPUs. According to the standard schemes in prior studies [16, 17], we evaluate the final checkpoints with 20 rollout trials for each task, repeated across three distinct random seeds to mitigate stochastic variability. In Tab 1, we report the average success rates for all tasks.

Quantitative analysis. As illustrated in Tab. 1, DeepVision-VLA achieves a mean success rate of 83% across 10 RLBench manipulation tasks, markedly surpassing all prior baseline methods. Specifically, our approach achieves substantial leads of 9%, 18%, and 22% over prevailing baselines such as HybridVLA (74%), $\pi_{0.5}$ (65%), and CogACT (61%), respectively. Beyond the aggregate metrics, DeepVision-VLA attains the highest performance in 8 out of the 10 tasks, underscoring its exceptional robustness and execution stability when handling heterogeneous manipulation skills. This performance surge primarily stems from our proposed Vision-Language Mixture-of-Transformers (VL-MoT) architecture coupled with the Action-Guided Visual Pruning (AGVP) strategy. As established in our layer-wise analysis, conventional VLA models typically suffer from “visual drift,” where the sensitivity to task-relevant visual tokens progressively decreases in deeper LLM layers. By introducing DINOv3 as a dedicated Vision Expert and asymmetrically injecting its multi-scale features directly into these attenuated deeper layers, VL-MoT effectively compensates for the decay of visual signals.

Crucially, this superior task performance does not compromise real-time control efficiency. Powered by the AGVP strategy, our framework leverages robust, unattenuated attention priors from shallow VLA layers to explicitly filter out redundant background information. This targeted pruning mechanism enables the Vision

Expert to process higher-resolution inputs and lock onto primary manipulation entities while maintaining a high-frequency inference speed. Comparative attention heatmaps confirm the synergy of these designs: while baseline models exhibit scattered and unstructured attention in deeper layers (e.g., failing to focus on the target objects during the “Sweep to dustpan” task), DeepVision-VLA maintains highly concentrated attention patterns that tightly align with physical object boundaries. This precise visual alignment at the perceptual source not only overcomes the detachment of deep features but also provides reliable spatial-geometric guidance for downstream action generation without incurring massive computational overhead.

4.2 Ablation Study

To validate the key designs of DeepVision-VLA, we conduct comprehensive ablation experiments on RLBench 10 manipulation tasks, using the same training and evaluation settings as shown in Sec. 4.1.

Importance of VL-MoT Architecture. We analyze the contribution of the VL-MoT design by exploring different paradigms for integrating Vision Expert features. Across 10 evaluation tasks, the vanilla VLA baseline achieves a 66% success rate. Directly concatenating the DINOv3 visual features before the Large Language Model (LLM), a common approach in early OpenVLA models, drops the average success rate substantially to 59%. Using DINOv3 as an intermediate layer aligner yields a modest 67.4% success rate. In contrast, our proposed Vision-Language Mixture-of-Transformers (VL-MoT) framework, which selectively couples the Vision Expert with the LLM backbone by injecting visual tokens directly into the deeper VLA layers, achieves a superior 73% success rate under the same conditions. This demonstrates that our targeted integration scheme is the most effective paradigm for overcoming visual drift.

Vision Foundation Model Feature Selection. We further investigate the impact of different multi-scale feature selection strategies from the Vision Expert across 10 tasks. Extracting features exclusively from the first 16 layers of DINOv3 results in a 66% success rate. Extracting from the back 16 layers improves this significantly to 77.2%. Uniformly sampling 16 layers across the entire DINOv3 model yields a robust 71%. Interestingly, repeatedly injecting features strictly from the very last layer of DINOv3 achieves an exceptional 83.2% success rate, indicating that high-level semantic cues from the Vision Expert provide the strongest supplementary guidance for deep VLA layers. Furthermore, substituting DINOv3 with a SigLIP encoder (sampling its last layer) only achieves 67% on 4 core tasks, confirming the necessity of a dedicated, high-resolution Vision Expert.

Guidance Mechanisms for Action-Guided Vision Pruning. To evaluate the effectiveness of the AGVP strategy, we compare different mechanisms for generating the visual attention guidance map used for pruning on 4 core tasks. The baseline model scores 65.5%. Using the DINOv3 CLS token to guide visual pruning provides no improvement, maintaining a 65.5% success rate. Relying on task instruction-to-vision attention improves performance to 84%. However, utilizing our proposed action-to-vision attention map natively from the shallow VLA layers achieves the highest

Table 1: Comparison of DeepVision-VLA and baselines on RL Bench. All methods are trained in the multi-task setting [44], and we report mean success rates (S.R.).

Models	Close box	Close laptop lid	Toilet seat down	Sweep to dustpan	Close fridge	Phone on base	Umbrella out	Frame off hanger	Wine at rack	Water plants	Mean S.R. ↑
OpenVLA	0.60	0.35	0.75	0.55	0.85	0.20	0.30	0.15	0.20	0.05	0.40
SpatialVLA	0.80	0.70	0.85	0.20	0.80	0.15	0.25	0.40	0.15	0.30	0.46
CogACT	0.90	0.80	0.95	0.50	0.85	0.50	0.55	0.45	0.30	0.25	0.61
CoT-VLA	0.95	0.75	1.00	0.80	0.65	0.50	0.40	0.50	0.55	0.50	0.66
$\pi_{0.5}$	0.90	0.95	0.85	0.75	1.00	0.05	0.10	0.80	0.75	0.35	0.65
HybridVLA	0.85	0.95	1.00	0.90	1.00	0.50	0.50	0.70	0.50	0.50	0.74
QwenVLA-OFT											
DeepVision-VLA	1.00	0.95	1.00	0.95	0.95	0.75	0.65	0.70	0.85	0.50	0.83

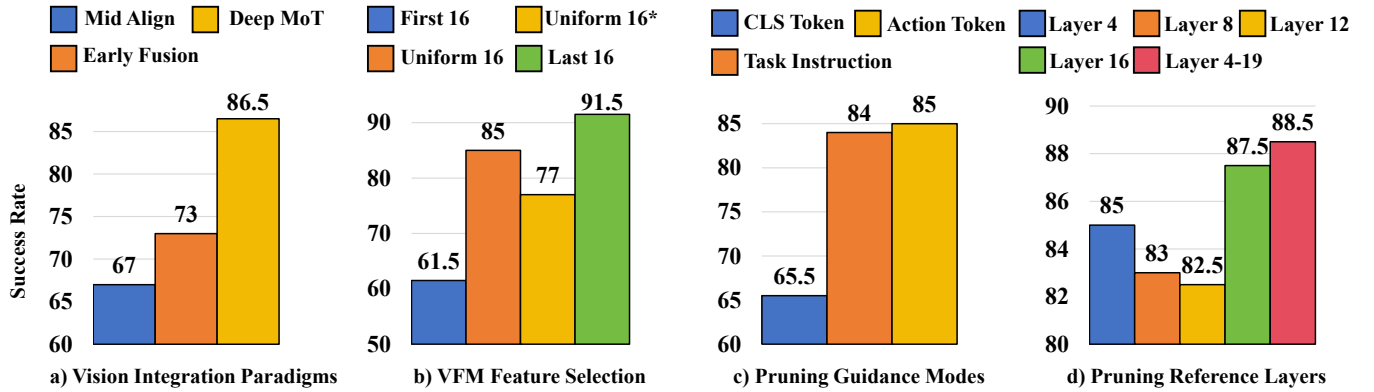


Figure 4: Ablation studies. (a) Comparison of different paradigms for integrating Vision Expert features into the VLA backbone. (b) Evaluation of different multi-scale feature selection strategies from the Vision Expert. (c) Comparison of different guidance mechanisms for generating the visual pruning map in AGVP. (d) Analysis of different shallow-layer references used to compute the action-to-vision attention map for pruning. Results are reported as average success rates across 10 RL Bench tasks.

Table 2: Comparison across real-world manipulation tasks.

Models	Stack coke cans	Write letter ‘S’	Pick fruit to the plate		Pour coke to bottle		Avg
			Step 1	Step 2	Step 1	Step 2	
$\pi_{0.5}$	0.65	0.95	0.75	1.00	1.00	0.70	0.842
OpenVLA-OFT	0.50	0.85	0.70	0.80	0.75	0.70	0.717
Qwen-OFT (baseline)	0.50	0.80	0.85	0.90	0.70	0.70	0.742
Ours: Qwen-OFT (DeepVision Improved)	0.65	0.95	0.95	0.95	1.00	1.00	0.917

performance at 85%, proving it to be the most reliable indicator for identifying task-relevant visual regions.

Impact of Reference Layers on AGVP. Finally, we analyze which specific layer’s action-to-vision attention map provides the best guidance for pruning on the 4 core tasks. Using the attention map from the 4th layer achieves an 85% success rate. Shifting the reference layer to the 8th layer causes a notable performance drop

to 69%. However, utilizing the attention maps from the 12th and 16th layers yields success rates of 82.5% and 87.5%, respectively.

4.3 Real-World Experiment

Data Collection. We evaluate DeepVision-VLA on four complex dual-arm real-world manipulation tasks using Franka Research 3 robot setups. As shown in Fig. xxx, we equip each robot arm with

an Intel RealSense D455 camera to capture wrist-view RGB observations, complemented by an exterior D455 camera providing the third-person perspective. The evaluated tasks encompass 1) *picking up a banana and a carrot*, 2) *stacking a Coke can*, 3) *pouring Coke*, and 4) *writing on a whiteboard*. To provide a more comprehensive assessment of model performance, we decompose the evaluation metrics for multi-stage tasks. Specifically, the picking task yields two separate success scores for the banana and the carrot, and the pouring task is evaluated independently on the initial grasp and the subsequent pouring action. In contrast, the Coke stacking and whiteboard writing tasks are assessed using a single comprehensive success score for the complete execution. More details are provided in Appendix xxx.

Training and evaluation details. The training protocol for the DeepVision-VLA remains the same as in the simulation. However, the real-world experiments differ by utilizing a dual-arm robotic setup that captures multi-view visual observations from two wrist cameras and a third-person camera. We compare our method against three state-of-the-art VLA model baselines including $\pi_{0.5}$ [6], OpenVLA-OFT [24], and QwenVLA-OFT [47]. These two OFT-based models specifically leverage parallel decoding and action chunking to enhance execution efficiency and action generation. For a fair evaluation, all baselines share the identical camera setup, with each task undergoing 20 rollouts under consistent test conditions.

Quantitative and qualitative analysis. Tab. 2 shows that DeepVision-VLA achieves the highest overall performance across all real-world evaluations with an average success rate of 91.7%, effectively surpassing the strong baselines including $\pi_{0.5}$ (84.2%), Qwen-OFT (74.2%), and OpenVLA-OFT (71.7%), which demonstrates the highly robust physical interaction and dynamic manipulation capabilities of our DeepVision-VLA. For the two single-step tasks evaluated with a comprehensive score, DeepVision-VLA attains success rates of 65% for stacking Coke cans and 95% for writing the letter ‘S’. The whiteboard writing task particularly highlights the strengths of our architecture, as the robot must precisely draw instructed shapes requiring continuous, high-precision spatial tracking. While conventional VLA models often suffer from visual drift in deeper network layers during such tasks, our dedicated Vision Expert and VL-MoT design effectively overcome this limitation. By asymmetrically injecting multi-scale visual features—ranging from low-level spatial textures to high-level semantic cues—directly into the deeper layers, our model compensates for visual signal attenuation and ensures accurate end-effector trajectory following.

Furthermore, DeepVision-VLA excels in multi-stage tasks evaluated via granular sub-stage metrics. During the “Pick fruit to the plate” task, it maintained a consistent 95% success rate across both Step 1 and Step 2. More notably, in the “Pour Water” task, our model achieved perfect 100% success rates across both stages, outperforming the strongest baseline ($\pi_{0.5}$), which dropped to 70% in the second step. These multi-stage tasks necessitate strict visual focus on object boundaries and relative positioning. Powered by the Action-Guided Visual Pruning (AGVP) strategy, our model leverages robust, unattenuated attention priors from shallow VLA layers to compute a saliency map that prunes redundant background information. This targeted pruning guides the Vision Expert to lock onto critical

manipulation entities with minimal computational overhead, providing the highly reliable spatial and geometric guidance required to ensure the success of complex, multi-stage manipulation tasks.

5 Conclusion

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